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Python Programming - 2301CS404

Lab - 5

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: `Python3@Lab` → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: `HELLO123` → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: `hi@#9` → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

In [21]:
```
s = input("Enter String")
lw = 0
up=0
d=0
sp=0

for i in s:
    if i.islower():
        lw+=1
    elif i.isupper():
        up+=1
    elif i.isdigit():
        d+=1
    else:
        sp+=1

print("Lowercase :",lw)
print("Uppercase:" ,up)
print("Digits :",d)
print("Special Character :",sp)
```

```
Lowercase : 7
Uppercase: 2
Digits : 1
Special Character : 1
```

2) Reverse Each Word

- Input: `Python is Easy` → `nohtyP si ysaE`
- Input: `Hello World` → `olleH dlroW`
- Input: `Learn Python Fast` → `nraeL nohtyP tsaF`

In [3]:
```
s1 = input("Enter your String")
res=""
li = s1.split()
print(li)
for i in li:
    i = i[::-1]
    res+=i + " "

print(res)
```

```
['Hello', 'World']
olleH dlroW
```

3) Case Pattern Identifier

- `PYTHON` → Uppercase
- `python` → Lowercase
- `Python Programming` → Title Case

In [45]:
```
s = input("Enter your String:")

if s.isupper():
        print(f"{s} Uppercase")
elif s.islower():
        print(f"{s} Lowercase")
elif s.istitle():
        print(f"{s} Title Case")
```

```
python Lowercase
```

4) First and Last Occurrence Finder

- `programming` , `g` → First:3 Last:10
- `banana` , `a` → First:1 Last:5
- `hello` , `z` → Not found

In [4]:
```
s = input("Enter your String:")
ch = input("Enter Character:")
if s.find(ch)!=-1 and s.rfind(ch)!=-1:
    print(f"First Occurence:{s.find(ch)},Last Occurence:{s.rfind(ch)}")
else:
    print("Not Found")
```

```
First Occurence:3,Last Occurence:10
```

5) Word Frequency Counter (ignore the case)

- `Python is easy and Python is powerful` → python: 2
- `Java is popular but java is verbose` → java: 2
- `C is fast` → c:1

In [5]:
```
s = input("Enter String:").lower()
ch = input("Enter Character:")
print(ch+":",s.count(ch))
```

```
python: 2
```

6) Remove Extra Spaces

- `  Python   is   fun` → `Python is fun`
- `  Hello   World` → `Hello World`
- `NoExtraSpace` → `NoExtraSpace`

In [13]:
```
s = input("Enter String:")
final_str = " ".join(s.split())
print(f"Final String:{final_str}")
```

```
['Python', 'is', 'fun']
Final String:Python is fun
```

7) Prefix and Suffix Removal

- `unhappy.txt` → `happy`
- `pretest.py` → `test`
- `unwanted.doc` → `wanted`

In [72]:
```
s = input("Enter String:")
prefix = input("Enter Prefix:")
suffix = input("Enter Suffix:")

s = s.removeprefix(prefix).removesuffix(suffix)
print(f"Final String:{s}")
```

```
Final String:happy.
```

8) Replace Vowels

- `Programming` → `Pr*gr*mm*ng`
- `Education` → `*d*c*t**n`
- `Sky` → `Sky`

In [77]:
```
s = input("Enter String:")
vowel="aeiouAEIOU"
for i in vowel:
    s = s.replace(i,"*")
print(s)
```

```
*d*c*t**n
```

9) String Compression

- `aaabbcccc` → `a3b2c4`
- `xxxyyyz` → `x3y3z1`
- `abcd` → `a1b1c1d1`

In [14]:
```
s = input("Enter string: ")

result = ""
count = 1

for i in range(len(s)):
    if i < len(s) - 1 and s[i] == s[i + 1]:
        count += 1
    else:
        result += s[i] + str(count)
        count = 1

print("Compressed string:", result)
```

```
Compressed string: a3b2c4
```

10) Toggle Case (without using str.swapcase())

- `PyThOn` → `pYtHoN`
- `HELlo` → `helLO`
- `Python` → `pYTHON`

In [90]:
```
s = input("Enter String:")
ans = ""
for i in s:
        if i.islower():
            ans += i.upper()
        elif i.isupper():
            ans += i.lower()
print(ans)
```

```
pYtHoN
```

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- `User123` → Valid
- `123User` → Invalid
- `Us@12` → Invalid

In [18]:
```
un = input("Enter username: ")
```

```
if un.isalnum() and not un[0].isdigit() and len(un) >= 6:
    print("Valid Username")
else:
    print("Invalid Username")
```

```
Invalid Username
```

12) Palindrome Checker (ignore the case)

- `Madam` → Palindrome
- `Level` → Palindrome
- `Python` → Not Palindrome

In [15]:
```
s = input().lower()
if s == s[::-1]:
        print("Palindrome")
else:
    print("Not Palindrome")
```

```
Palindrome
```

13) Longest Word Finder

- `Python programming is interesting` → programming
- `I love coding` → coding
- `Data structures and algorithms` → structures

In [19]:
```
s = input("Enter sentence: ")

words = s.split()
longest = ""

for word in words:
    if len(word) > len(longest):
        longest = word

print("Longest word:", longest)
```

```
Longest word: programming
```

14) Case Conversion

- `python programming`
  - Upper: PYTHON PROGRAMMING
  - Lower: python programming
  - Title: Python Programming
  - Capitalize: Python programming

In [94]:
```
s = input()
print("Upper:",s.upper())
print("Lower:",s.lower())
print("Title:",s.title())
print("Capitalize:",s.capitalize())
```

```
Upper: PYTHON PROGRAMMING
Lower: python programming
Title: Python Programming
Capitalize: Python programming
```

15) Custom Split

- `apple,banana,grape`
- `10|20|30`
- `A-B-C-D`

In [99]:
```
s = input("Enter String:")
replace = input('Enter replace:')
li = s.split(replace)
print(li)
```

```
['apple', 'banana', 'grape']
```

Thank You

In [ ]: